



Problem solving with factors, multiples, squares and cubes

1 The product of two repeated integers is called a _____. Tick **1** correct answer

- ☐ multiple
- ☒ square number
- ☐ cube number
- ☐ factor

2 True or false? All integers are square numbers. Tick **1** correct answer

- ☐ True, you can square any integer.
- ☒ False, not all integers can be written as the product of two repeated integers.

3 What are the dimensions of the largest square you can make with 90 counters? Tick **1** correct answer

- ☐ 7 by 7
- ☐ 8 by 8
- ☒ 9 by 9
- ☐ 10 by 10

4 Which of the following is both a square number and a cube number? Tick **1** correct answer

- ☐ 4
- ☐ 8
- ☐ 36
- ☒ 64



5 Which of the following is not written as a product of prime factors? Tick **1** correct answer

☐ 2^3

☐ $3^4 \times 7$

☒ $5^2 \times 9$

☐ $11^3 \times 13 \times 19^2$

6 Which of the following are a correct answer to a string of consecutive numbers that satisfies: even, prime, composite, square Tick **2** correct answers

☐ 20, 21, 23, 24☒ 22, 23, 24, 25☒ 6, 7, 8, 9☐ 12, 13, 14, 15